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SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

(Established under section 3 of the UGC Act 1956)

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Semester: III **Year:** AY 26-27

Subject Title: Operating Systems Lab	
EXPERIMENT No: 3	
TITLE: Basic Shell script for File operation	**DoP**: 27-07-1026

Aim: Implement and demonstrate a shell script on file operations

Learning Outcome: 1. To understand the shell script for file operation
2. To demonstrate the shell script for file operation

Hardware/Software:

This experiment is performed using Ubuntu OS in a virtualized environment within Windows 11. The software used is:

- VMware Workstation 17 Player (Free Virtualization Tool)
- Ubuntu 22.04.1 LTS ISO file
- VMware Tools installed on the virtual machine

Theory:

A shell script is a sequence of system commands stored in a plain text file that is interpreted and executed sequentially by the command-line shell. In the context of operating systems, shell scripting serves as a powerful mechanism to automate routine administrative tasks and manage file systems without needing to manually type each instruction into the terminal. By utilizing a shell interpreter, such as /bin/sh, the script acts as an intermediary, interacting directly with the operating system to handle basic input/output operations, manipulate file attributes, and streamline repetitive system tasks.

The implemented program theoretically demonstrates the complete lifecycle of file management through an interactive workflow. It dynamically accepts user input during runtime to perform a series of operations on a target file, ranging from initial creation and standard data entry to advanced text transformations like alphabetical sorting, case conversion, and pattern matching. By redirecting input streams and passing data between utilities, the script highlights how simple, modular operating system programs can be combined to achieve complex file manipulation, ultimately concluding with the secure deletion of the system resource.

Command Details :

1. `echo`

The `echo` command is one of the most fundamental and frequently used built-in commands in Linux and Unix-like operating systems. Its primary purpose is to display a line of text or string onto the standard output, which is typically the terminal screen. This command is highly versatile and is essential in shell scripting for outputting status messages, displaying variable values, or providing interactive prompts to users during script execution.

Beyond simple text display, `echo` can be combined with escape sequences to format output, such as adding newlines or tabs. It also plays a crucial role in file redirection, where instead of printing to the screen, its output can be directed into a file using the `>` or `>>` operators. This makes it an indispensable tool for generating logs and dynamically creating files within scripts.

Syntax : echo [string/text]

Example : echo "Enter name"

2.`read`

The `read` command is a built-in shell utility used to take input from the user via the keyboard (standard input) during the execution of a script. When the interpreter encounters the `read` command, it pauses the script's execution and waits for the user to type a line of text and press the Enter key.

Once the input is received, the `read` command assigns the typed string to a specified variable. This allows shell scripts to be interactive, dynamic, and responsive to user choices, rather than operating solely on hardcoded values.

Syntax : read [variable_name]

Example : read name

3. `tr`

The `tr` (translate) command is a command-line utility used for translating, squeezing, or deleting characters from standard input. Unlike many other text processing tools, `tr` does not directly read from files; instead, it accepts input through standard input via piping (using the `|` symbol) or redirection. It operates on a character-by-character basis, mapping characters from a specified first set to corresponding characters in a second set.

In shell scripting, `tr` is exceptionally useful for rapid text transformations. A common use case is changing text casing—mapping the set of all lowercase letters (`[a-z]`) to the set of all uppercase letters (`[A-Z]`). Because it relies on data streams, it is almost always used in combination with commands like `cat` or `echo`.

Syntax : tr [set1] [set2]

Example : cat hello.txt | tr [a-z] [A-Z]

4. `grep`

The `grep` command (Global Regular Expression Print) is an essential search utility used to scan plain-text data for lines that match a specific regular expression or keyword. When `grep` finds a match within a file, it extracts and prints the entire line containing that matched string to the standard output.

This utility is highly valued for its ability to quickly filter through massive datasets to find specific information. In shell scripts, it is often used interactively to allow users to search for distinct patterns or to verify if a specific configuration string exists within a file before the script proceeds.

Syntax : grep [pattern] [filename]

Example : grep "$key" "$fname"

5. `mv`

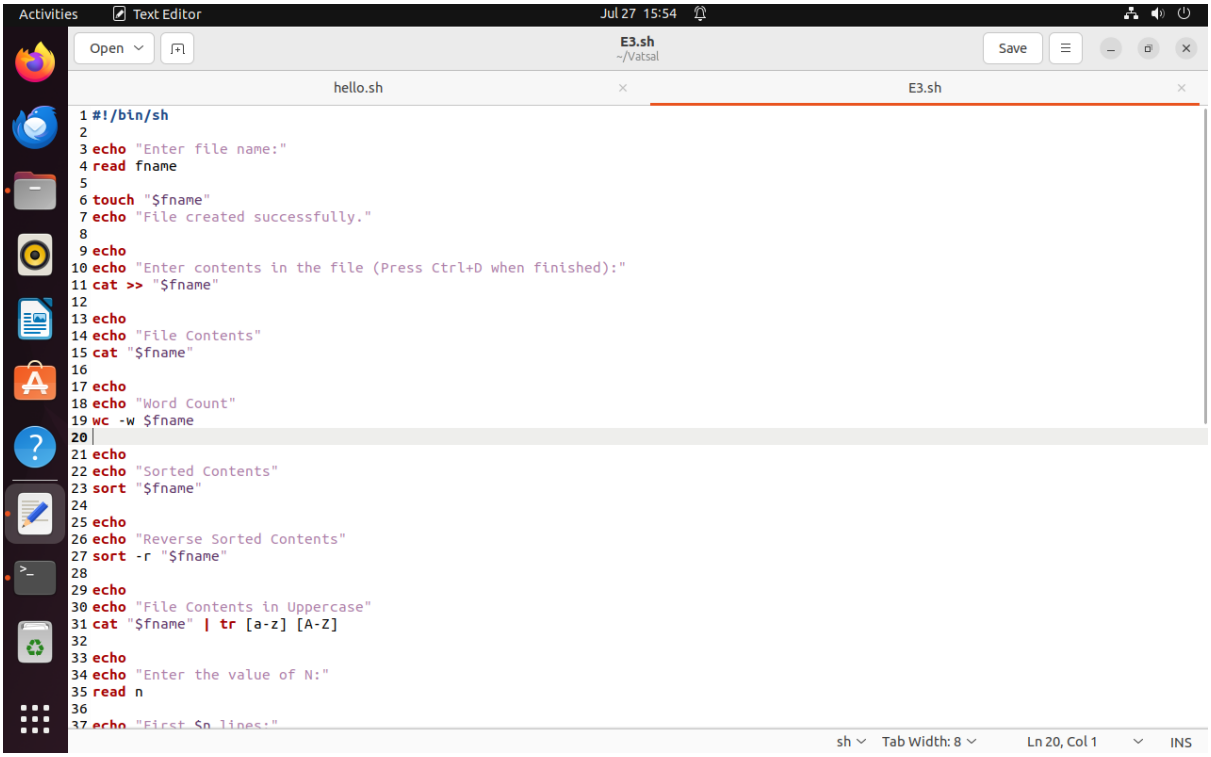
The `mv` (move) command serves a dual purpose in Unix-like operating systems: it is used to move files and directories from one location to another, and it is also the standard command used to rename them. Unlike copying a file, using `mv` does not create a duplicate; it alters the existing file's path or name within the file system hierarchy.

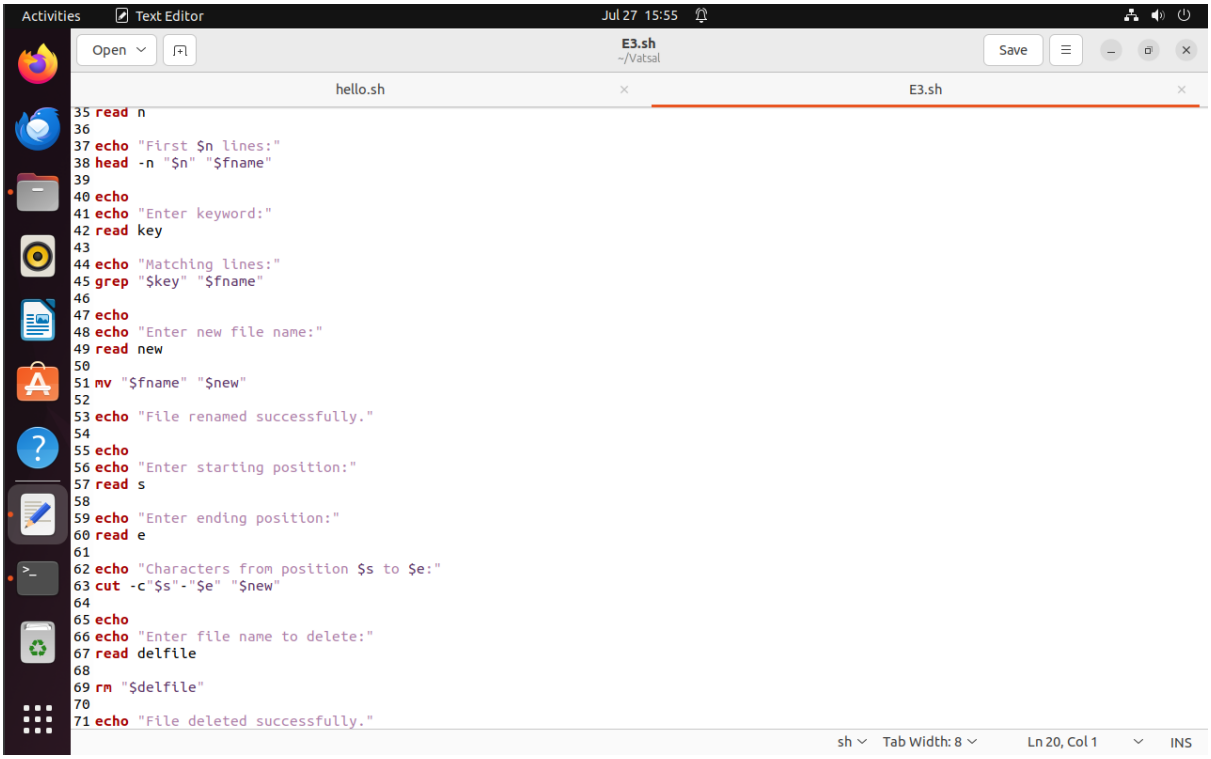
When moving a file to a new directory, the original filename is preserved unless a new name is specified. When used strictly for renaming—as seen in your script—both the source and destination paths remain in the same directory, but the file identifier is updated from the old name to the new name.

Syntax : mv [source_file] [destination_file]

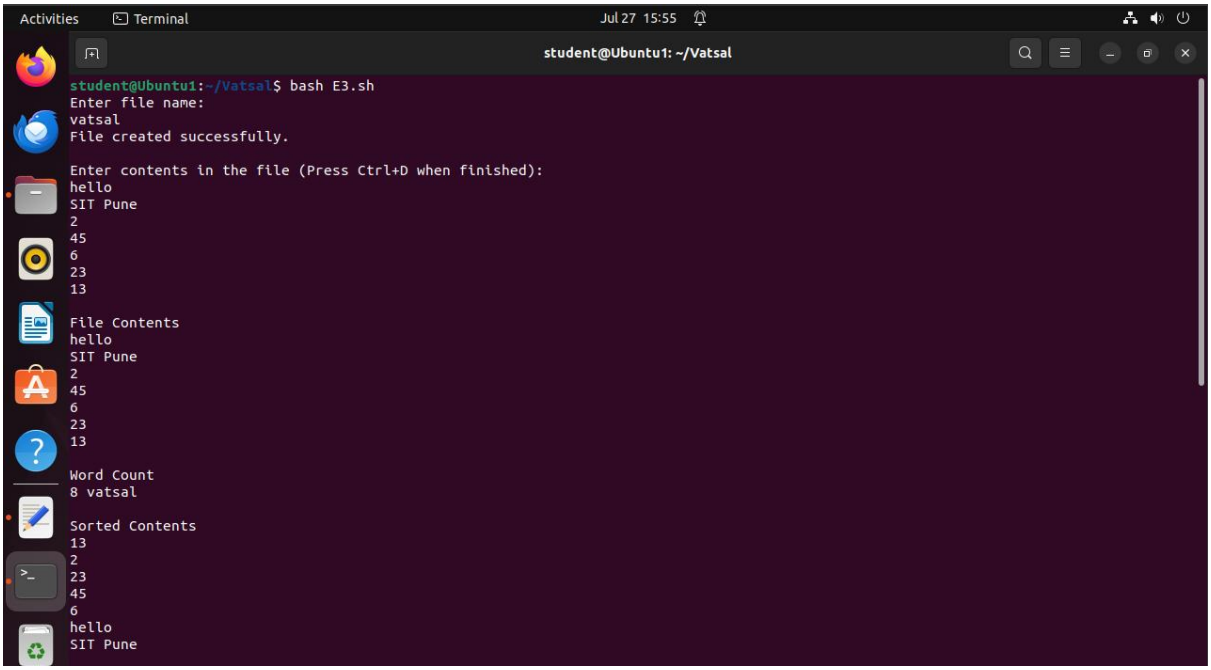
Example : mv "$fname" "$new"

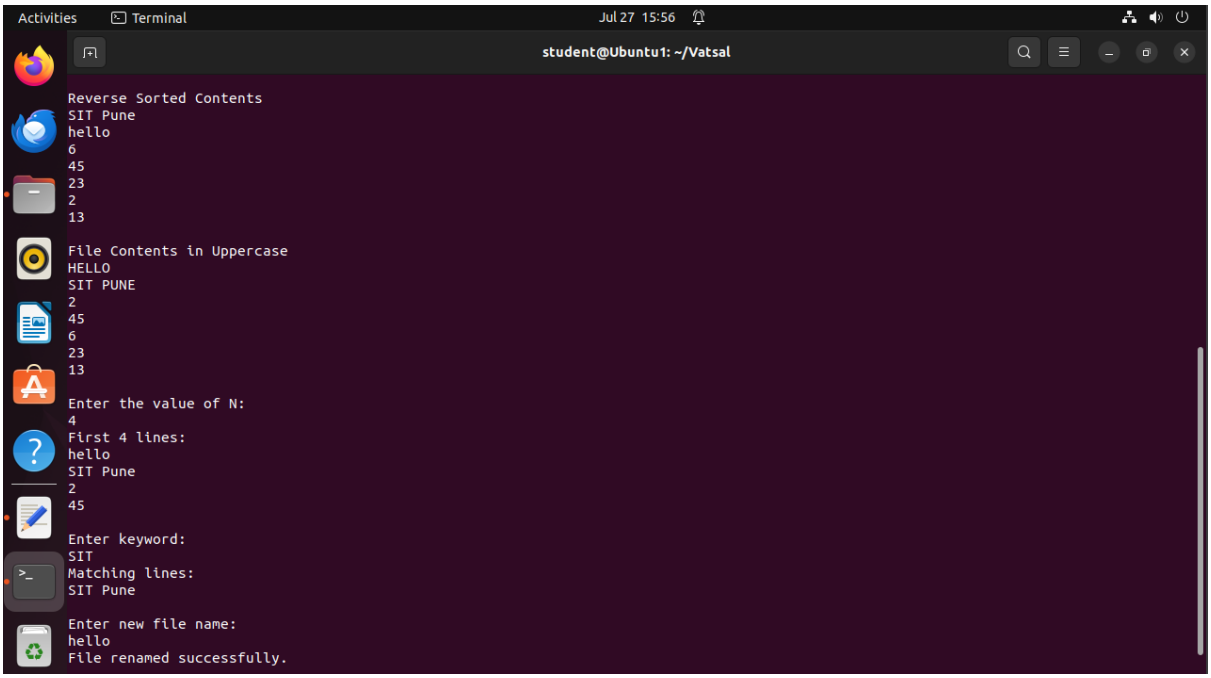
Program:

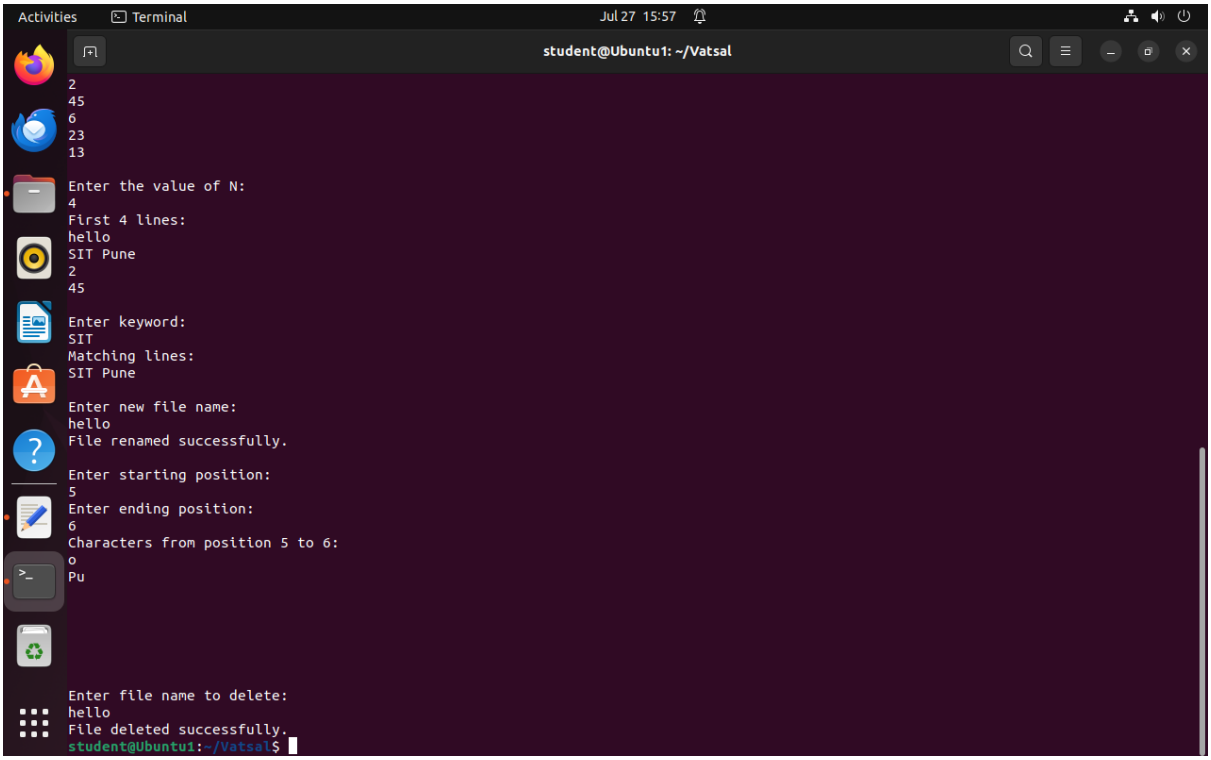

Activities
Text Editor
Jul 27 15:54
Open
E3.sh
~/Vatsal
Save
hello.sh
E3.sh
1 #!/bin/sh
2
3 echo "Enter file name:"
4 read fname
5
6 touch "$fname"
7 echo "File created successfully."
8
9 echo
10 echo "Enter contents in the file (Press Ctrl+D when finished):"
11 cat >> "$fname"
12
13 echo
14 echo "File Contents"
15 cat "$fname"
16
17 echo
18 echo "Word Count"
19 wc -w $fname
20
21 echo
22 echo "Sorted Contents"
23 sort "$fname"
24
25 echo
26 echo "Reverse Sorted Contents"
27 sort -r "$fname"
28
29 echo
30 echo "File Contents in Uppercase"
31 cat "$fname" | tr [a-z] [A-Z]
32
33 echo
34 echo "Enter the value of N:"
35 read n
36
37 echo "First $n lines:"
sh
Tab Width: 8
Ln 20, Col 1
INS


Activities
Text Editor
Jul 27 15:55
Open
E3.sh
~/Vatsal
Save
hello.sh
E3.sh
35 read n
36
37 echo "First $n lines:"
38 head -n "$n" "$fname"
39
40 echo
41 echo "Enter keyword:"
42 read key
43
44 echo "Matching lines:"
45 grep "$key" "$fname"
46
47 echo
48 echo "Enter new file name:"
49 read new
50
51 mv "$fname" "$new"
52
53 echo "File renamed successfully."
54
55 echo
56 echo "Enter starting position:"
57 read s
58
59 echo "Enter ending position:"
60 read e
61
62 echo "Characters from position $s to $e:"
63 cut -c"$s"-"$e" "$new"
64
65 echo
66 echo "Enter file name to delete:"
67 read delfile
68
69 rm "$delfile"
70
71 echo "File deleted successfully."
sh
Tab Width: 8
Ln 20, Col 1
INS

Output Screenshots:







Conclusion:

In this experiment, a shell script for basic file operations was successfully implemented and executed using the Ubuntu operating system within a virtualized environment. Through this script, various standard Linux commands were practically demonstrated, including file creation, text insertion, and data manipulation using utilities such as sort, tr, and cut. Additionally, the script successfully showcased file management and information retrieval techniques using grep, wc, head, mv, and rm. Overall, this experiment provided a hands-on understanding of how shell scripting can be utilized to automate routine tasks, manage system files, and process text data efficiently within a Linux environment.

Assessment Schemes:

Attendance	Discipline	Short oral	Correctness of Lab Report	Timely completion of Lab Report	Total marks (10)	Signature of Teacher with Date